Nolan Pierce

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Computer science student (Stanford, Visual Computing) pivoting toward hardware-software integration in robotics and manufacturing, with robotics/embedded systems coursework and Boeing experience in real-time 3D simulation.

Education

Stanford University — Stanford, CA
Bachelor of Science in Computer Science; GPA: 3.7 — *09/2023 – 06/2027*
Relevant Coursework: Introduction to Robotics (CS223A), Introduction to Electrical Engineering (ENGR40M), Computer Organizations and Systems (CS107), Deep Learning for Computer Vision (CS231N)

Technical Skills

Hardware & Robotics: Microcontroller programming (Arduino), closed-loop sensor/actuator control, servo & PWM actuation, serial communication (UART), circuit prototyping & debugging (ENGR40M), robotics fundamentals – kinematics, control (CS223A)
Languages: Python, C, C++, SQL
Simulation & Graphics: Unreal Engine 5 (Blueprint scripting, event dispatchers, widget/UI systems)
Computer Vision & ML: PyTorch, OpenCV, Convolutional Neural Networks (CNNs), Vision Transformers (ViT)
Databases: PostgreSQL, Supabase (Auth/RLS)
Frameworks & Tools: Node.js, Express.js, Git, Unix, gdb, REST APIs, JWT Authentication

Relevant Work Experience

Boeing Global Services — Seattle, WA
Systems Engineering Intern – Data & Analytics, VR Simulation & Modeling — *06/2026 – 09/2026*

- Engineered a virtual reality showroom in Unreal Engine 5 presenting employee-developed and patented safety and ergonomics solutions, extending a single physical "Dojo" showcase into a **scalable, site-independent environment**
- Architected interactive Blueprint systems – custom UI widgets, interface classes, event dispatchers, and parent/child blueprint hierarchies – to **streamline integration** of new showcase content into the existing simulation architecture
- Presented live demos of the environment to **Boeing leadership**

Mission Malama — Remote
Founding Engineer — *07/2025 – 09/2025*

- Designed and implemented backend architecture – REST API endpoints, database schema, and CRUD operations – using Node.js, Express.js, and Supabase (Postgres, Auth/RLS)
- Implemented JWT-based authentication and role-based access controls to ensure secure handling of sensitive user data

Washington Technology Student Association — Remote
Alumni Representative — *03/2023 – Present*

- Manage a network of 150+ alumni, facilitating mentorship programs, event planning, and volunteer recruitment

Candidate & Entertainment Manager — *03/2026 – Present*

- Serve on the interview committee, narrowing 15 applicants to a Top 12, then manage that candidate pool on-site through the state conference
- Plan and execute entertainment logistics for a 2,000+ attendee state conference across 3 event nights

Personal Projects

Distance-Triggered Servo Gate | *Arduino, C++, HC-SR04, SG90 servo*

- Built a closed-loop embedded control system: an ultrasonic sensor measures distance by pulse timing and commands a servo through PWM actuation when an object crosses a configurable threshold
- Validated the sensor subsystem over serial before actuator integration; full build, wiring, and debug completed in 1.5 hours

Multi-Agent Tracking & Trajectory Forecasting | *Python, PyTorch, OpenCV, Modal*

- Built an end-to-end perception-to-prediction pipeline for basketball video: SAM 3.1 multi-object player tracking (SportsMOT) feeding a residual, rule-conditioned LSTM in PyTorch that predicts 4 frames from an 8-frame window
- Matched a constant-velocity baseline (5.81 px median displacement error) and outperformed a plain LSTM on 10 of 12 test windows; trained on GPU via Modal, validating transfer across held-out clips

Leadership & Extracurriculars

Stanford Men's Rugby — Stanford, CA
President, Vice President — *05/2025 – Present*

- Led 40+ team members and managed administrative operations for the club